KESHAV NARAM

+1 (240) 486-6826 naramkeshav59@gmail.com keshav-naram Naramkeshav59 Portfolio

Skills

Machine Learning & AI	Deep Learning, Computer Vision, NLP, LLMs & Agents, RAG, Fine-tuning (PEFT / QLoRA), Speech Recognition
Frameworks & Libraries	PyTorch, Hugging Face Transformers, LangChain / LangGraph, scikit-learn, OpenCV, Whisper, Wav2Vec 2.0, ChromaDB, NumPy, Pandas
Programming Languages	Python, SQL, Bash
Data Engineering & Cloud	AWS (Lambda, API Gateway, SQS, ECS Fargate, S3, DynamoDB, CloudFormation), DuckDB, Docker, Kubernetes
MLOps & Tools	MLflow, Airflow, Jenkins, GitHub Actions, Git, Linux, Streamlit, Flask, pytest

Projects

ShuttleCast: AI Tactical Analyst for Pro Badminton — *LangGraph, PyTorch, AWS* Demo • 2026

- Most match footage has no stroke labels, so trained stroke-detection models (frozen DINOv2 + temporal CNN) on ShuttleSet to read raw video, reaching hit-detection F1 0.87 on a held-out match vs 0.37 for an audio baseline.
- Coaching notes read well but blamed the rally's winner; designed evals for a LangGraph ReAct agent and grounded its prompt in the outcome, lifting correct attribution from 62% to 94% in an A/B with decisions held fixed.
- Full matches outlast Lambda's 15-minute limit, so deployed an async pipeline (API Gateway, Lambda, SQS, Fargate) that streams notes per rally, surfacing the first in ∼5 min instead of ∼11; QLoRA took a 2B model to 100% valid output.

FHIRFlow: Making Hospital Lab Data Queryable — *FHIR, DuckDB, LLMs* 2026

- Hospital labs use local codes that standard queries miss; engineered a FHIR R4 to DuckDB pipeline with data-quality rules in SQL, exposing that 0% of MIMIC-IV's 107K lab results were findable by LOINC across 498 local codes.
- Manual LOINC mapping takes expert weeks and LLMs invent codes; developed a retrieval-grounded mapper that picks only retrieved codes or abstains, reaching 65% exact match vs 17.5% for rules and correctly coding 87% of lab rows.

FinSight: Cited Q&A over SEC 10-K Filings — *RAG, ChromaDB, Ollama/Groq* Demo • 2026

- Analysts can't trust answers they can't trace; built a hybrid RAG system (dense + BM25) over SEC 10-K filings that cites the exact source passage for every claim and diffs the same section across filing years.
- Standard RAG components are untested on financial text; benchmarked 9 chunking × retrieval strategies and found cross-encoder reranking lowered answer similarity (0.79 vs 0.85) while adding latency on 10-K text.

Experience

Software Intern – AI/ML — *WyzMindz Solutions Pvt. Ltd. (Remote)* Jul 2022 – Dec 2022

- Call audio is heavy with silence and hold noise; implemented real-time voice activity detection that removed ∼35% of non-speech audio before recognition, cutting speech-to-text compute by a similar share.
- Pretrained recognizers struggled with call-center noise; fine-tuned Wav2Vec 2.0 on in-domain audio and integrated Kaldi/Fairseq recognizers, reducing word error rate by ∼30% relative to the base model.
- Agents needed transcripts during live calls; delivered a Flask streaming API combining VAD and speech-to-text that showed text on screen within ∼500 ms of speech.

Education

University of Maryland, College Park May 2026
Master of Science in Applied Machine Learning GPA 3.73 / 4.00

Indian Institute of Technology, Dharwad 2020 – 2024
Bachelor of Technology GPA 8.08 / 10.00

Certifications & Leadership

- DeepLearning.AI: Neural Networks and Deep Learning; Natural Language Processing Specialization.
- Academic mentor, IIT Dharwad (Aug 2022 – Aug 2023): guided 20+ undergraduates on course and career planning.